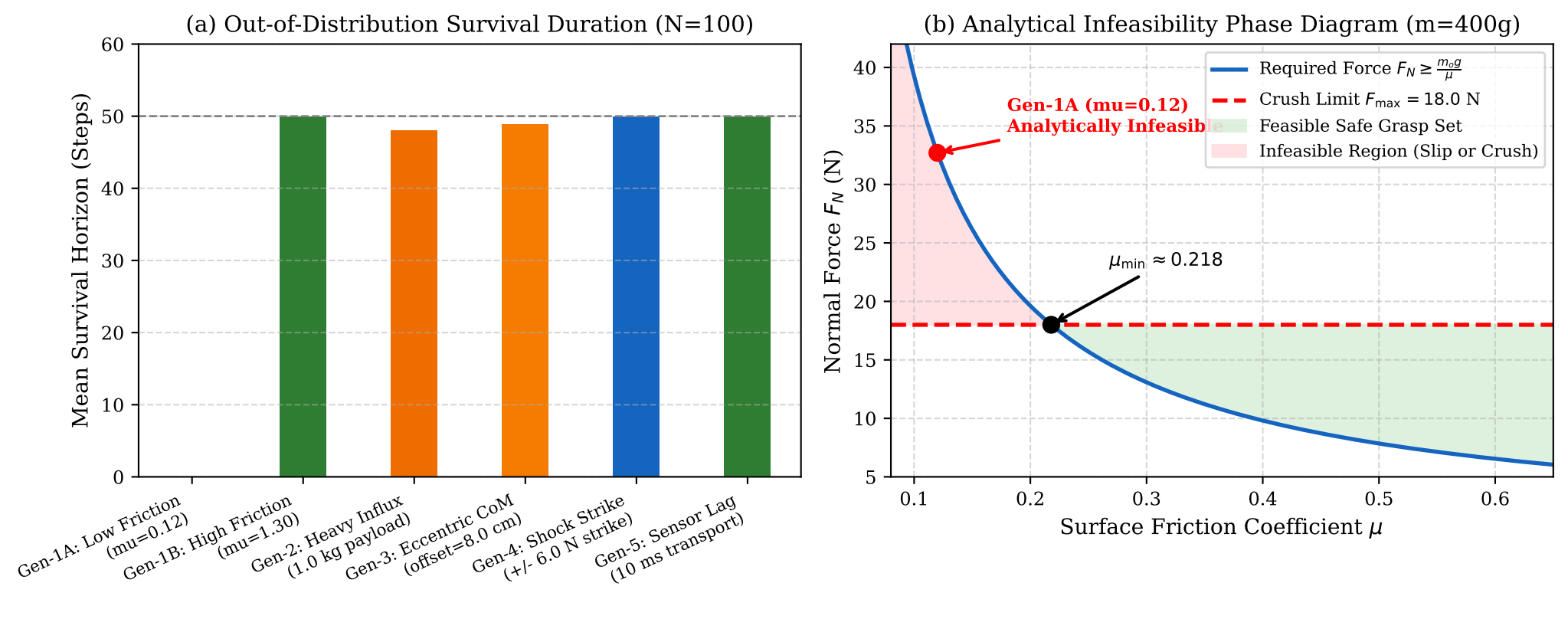
(a) Out-of-Distribution Survival Duration (N=100)
Mean Survival Horizon (Steps)
0
10
20
30
40
50
60
Gen-1A: Low Friction (mu=0.12)
Gen-1B: High Friction (mu=1.30)
Gen-2: Heavy Influx (1.0 kg payload)
Gen-3: Eccentric CoM (offset=8.0 cm)
Gen-4: Shock Strike (+/- 6.0 N strike)
Gen-5: Sensor Lag (10 ms transport)
(b) Analytical Infeasibility Phase Diagram (m=400g)
Normal Force $F_N$ (N)
Surface Friction Coefficient $\mu$
Required Force $F_N \geq \frac{m_o g}{\mu}$
Crush Limit $F_{max} = 18.0$ N
Feasible Safe Grasp Set
Infeasible Region (Slip or Crush)
Gen-1A (mu=0.12)
Analytically Infeasible
$\mu_{min} \approx 0.218$